

Project 2

Memory card game

- display 5 random numbers in box
- and let user give the correct sequence of that numbers in a input
- the game is 3 times
- if they score 0/3 display (may be next time)
- if they score 1/3 display (it was your choice)
- if they score 2/3 display (use this brain to figure out the world)
- if they score 3/3 display (shocking)

First, I created a React app in Terminal with the command `npx create-react-app memory-card-game`. Then removed all the unnecessary files that were not required.

I created numbers and setNumbers using `useState` to store the five randomly generated numbers.

Then, I created a `generateNumbers` function to generate five random numbers. Inside the function

I created an empty `newNumbers` array and used a for loop to generate 5 random numbers from 0 to 9 using `Math.floor(Math.random() * 10)`. I used `push()` to add the numbers to the array and `setNumbers(newNumbers)` to store them in the numbers state.

I created `input` and `setInput` to store the number entered by the user. I created `round` and `setRound` to keep track of the round and set the initial value as 1.

I created `score` and `setScore` to store the user's score and set the initial value as 0. I created `message` and `setMessage` to display the final message based on the user's score.

I created `correct` and `setCorrect` to display whether the answer entered by the user is Correct or Incorrect.

I created `showNumbers` and `setShowNumbers` to control whether the generated numbers should be displayed or hidden and set the initial value as `true`.

Then, I created a `checkAnswer` function to check the user's answer. I used `numbers.join("")` to convert the generated numbers into a single number string for comparison.

I created a `newScore` variable and assigned the current score to it so that I could immediately use the updated score when checking the final round.

I compared the user's input with the `correctAnswer`. If it matches, I increased the score by 1 and displayed `Correct`. Otherwise, I displayed `Incorrect`.

After checking the answer, if the round is 3, the game ends. I displayed a message based on the score: 3 = "Shocking", 2 = "Use this brain to figure out the world", 1 = "It was your choice", and 0 = "May be next time".

If the round is not 3, I increased the round by 1 using `setRound(round + 1)` and cleared the input using `setInput("")` for the next round.

Then, I used `useEffect` to generate new numbers and control the timer whenever the round changes.

I added `round` to the dependency array, so `useEffect` runs whenever the round changes. Inside it, I called `generateNumbers()` to create five new random numbers.

Then I created a timer using `setTimeout`. I used 3000 milliseconds, which is equal to 3 seconds. This means the generated numbers remain visible for 3 seconds.

After 3 seconds, I used `setShowNumbers(false)` to hide the numbers and display `?????`. I also used `clearTimeout(timer)` as a cleanup function to cancel the timer when needed.

`clearTimeout()` cancels the timer before it finishes, preventing the old timer from affecting the new round.

After Round 3, the button becomes `Restart Game`. Clicking it resets the game, and setting the round to 1 triggers `useEffect` to generate new numbers.

Results :-

Memory Card Game

Round 1/3

?????

Enter the Number

Check Answer

Score: 0/3

Memory Card Game

Round 2/3

?????

Enter the Number

Check Answer

Incorrect

Score: 0/3

Memory Card Game

Round 3/3

?????

Enter the Number

Restart Game

Correct

Score: 1/3

It was your choice

Challenges faced in this project:

Managing multiple states such as numbers, input, round, score, message, correct, and showNumbers.

Generating five random numbers for each round and storing them in an array.

Hiding the numbers after 3 seconds using setTimeout.

Using clearTimeout() to prevent the previous timer from affecting the next round.

Managing 3 rounds and updating the score correctly.

Making sure the UI updates correctly when the numbers, score, round, and answer status change.

After Round 3, the button becomes Restart Game.